



EE 671: VLSI Design

Course Project-1

Team 10

Team Members

S.No.	Name	Roll Number
1	Meet Jain	26M1164
2	Rajas Tatwawadi	26M1161
3	Abhineet Singh	26M1191
4	Uday Gupta	26X0040

Cells Assigned

S.No.	Cell	Description
Cell 1	buf	Buffer
Cell 2	dfxtp	Positive Edge D flip-flop
Cell 3	nand2b	2-input NAND (1st input inverted)

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1 Cell: buf

1.1 Circuit Diagram and Transistor Sizing

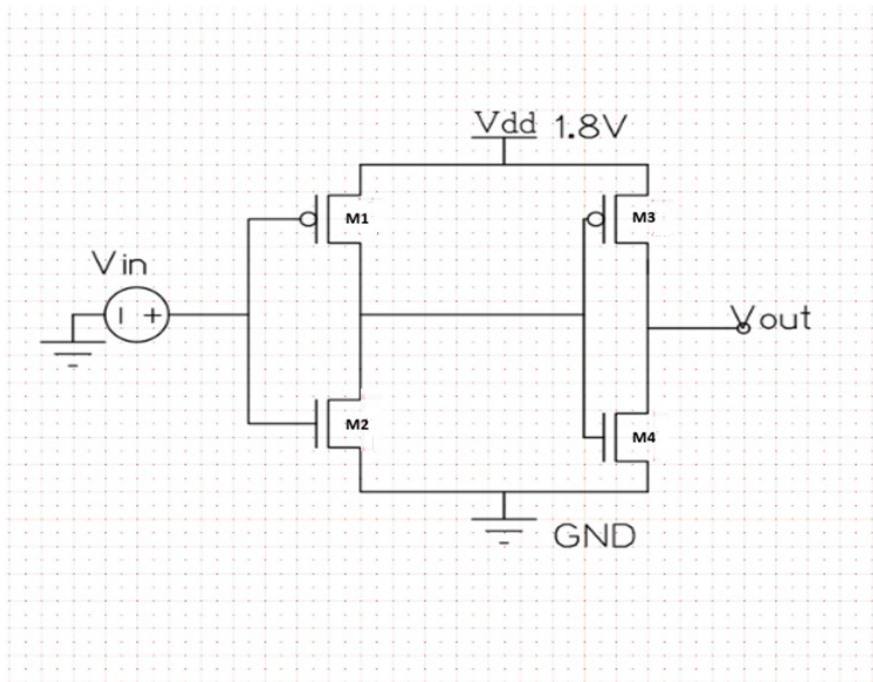


Figure 1: Circuit diagram of buf

Table 1: MOSFET widths and lengths for buf

Device	Type	W (μm)	L (μm)
M1	NMOS	0.42	0.15
M2	PMOS	1.23	0.15
M3	NMOS	0.42	0.15
M4	PMOS	1.23	0.15

1.2 Layout

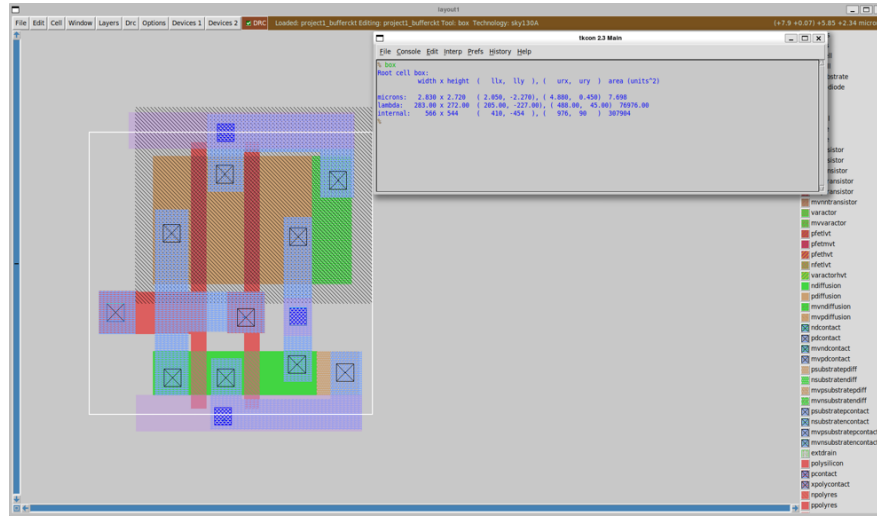


Figure 2: Layout of buf in Magic

Cell area: $WIDTH \times HEIGHT = 2.83\mu m \times 2.72\mu m$

1.3 PEX Netlist

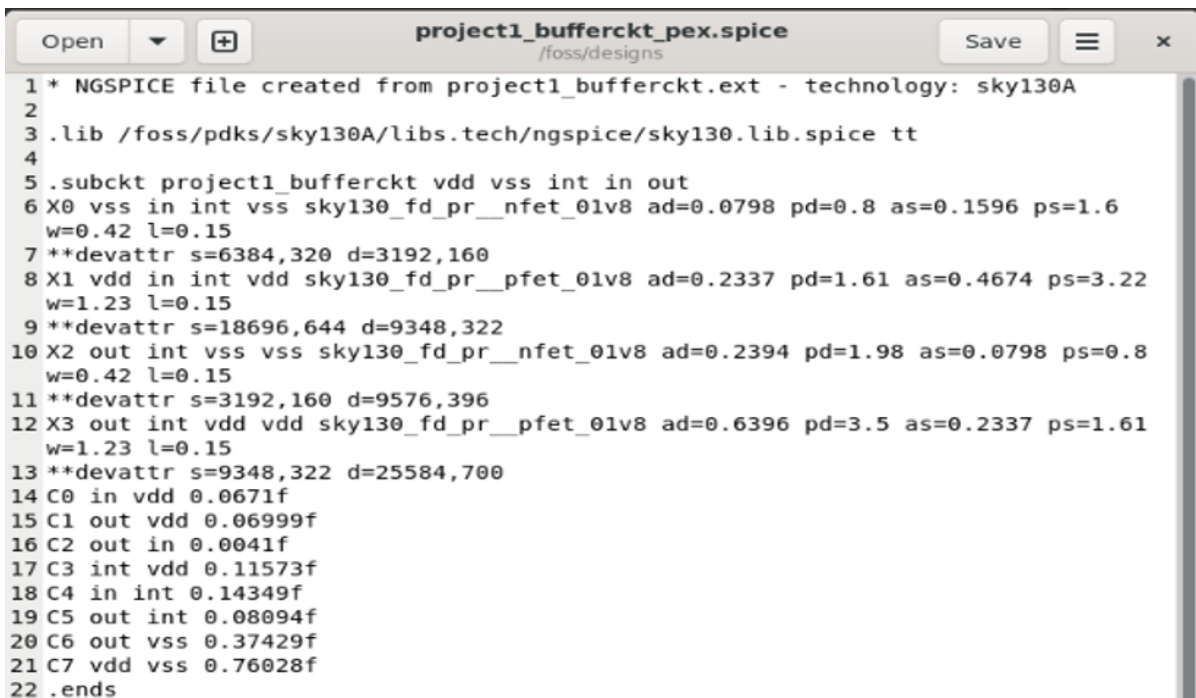


Figure 3: Screenshot of the PEX netlist of buf

1.4 DRC and LVS

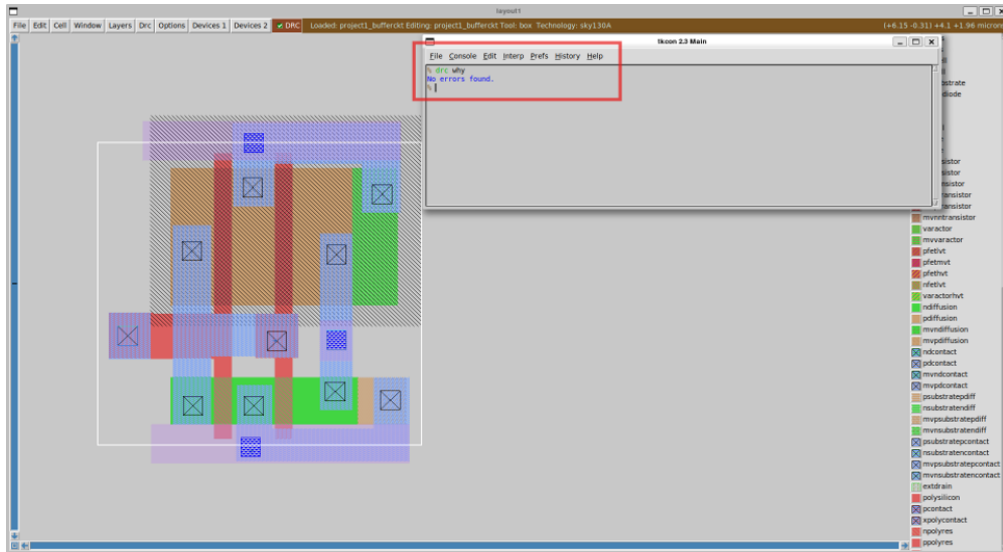


Figure 4: DRC clean screenshot for buf

```

Contents of circuit 1: Circuit: 'project1_bufferckt'
Circuit project1_bufferckt contains 4 device instances.
  Class: sky130_fd_pr__pfet_01v8 instances:  2
  Class: sky130_fd_pr__nfet_01v8 instances:  2
Circuit contains 5 nets.
Contents of circuit 2: Circuit: 'bufferckt'
Circuit bufferckt contains 4 device instances.
  Class: sky130_fd_pr__pfet_01v8 instances:  2
  Class: sky130_fd_pr__nfet_01v8 instances:  2
Circuit contains 5 nets.

Circuit 1 contains 4 devices, Circuit 2 contains 4 devices.
Circuit 1 contains 5 nets,    Circuit 2 contains 5 nets.

Final result:
Circuits match uniquely.
.
Logging to file "comp.out" disabled
LVS Done.
/foss/designs/project1/project1_buffer >

```

Figure 5: LVS clean screenshot for buf

1.5 Simulation Waveforms

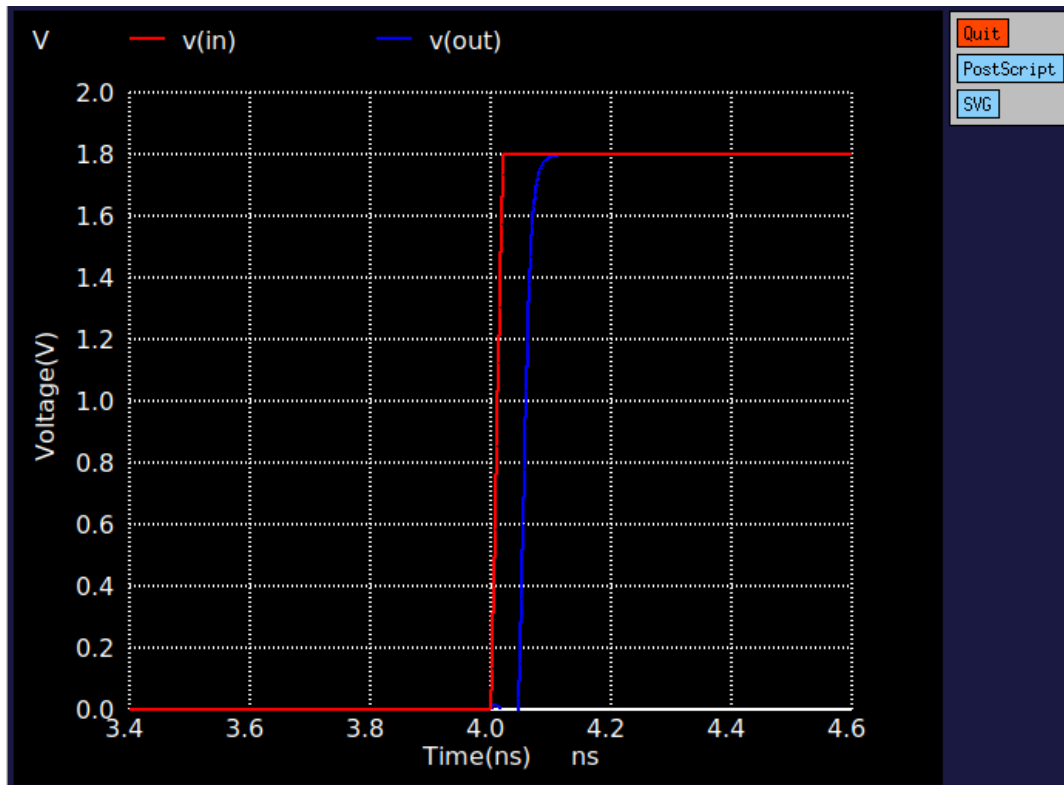


Figure 6: Propagation delay waveform (input, output) for buf

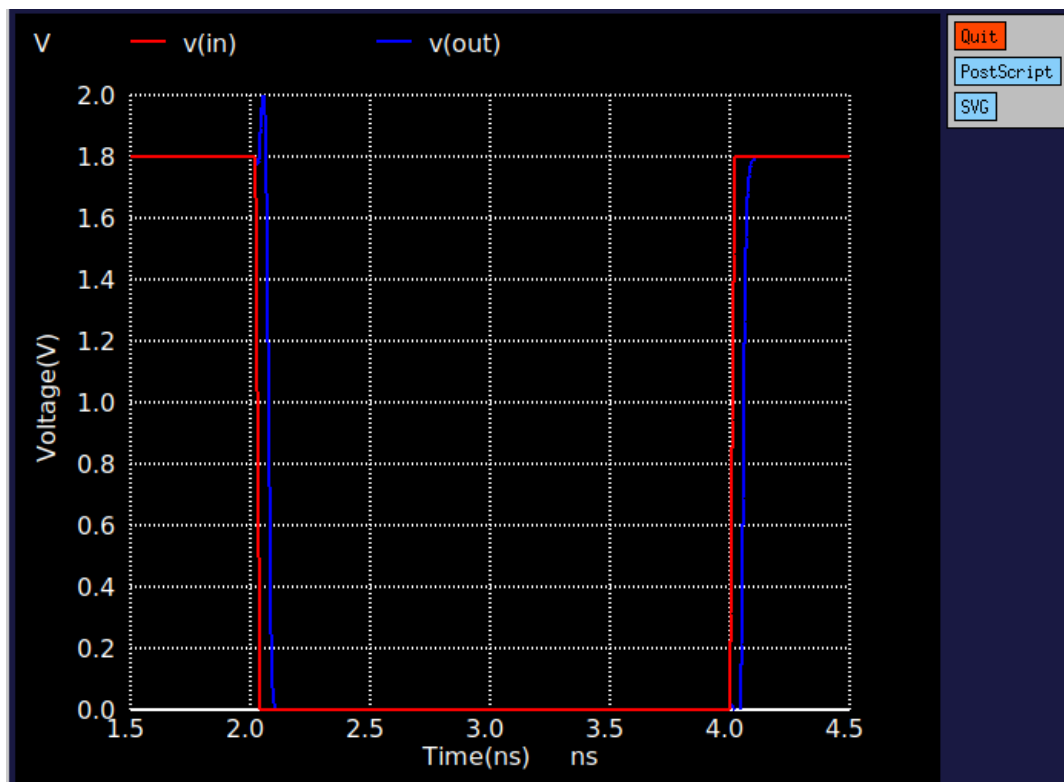


Figure 7: Rise/fall transition time waveform for buf

1.6 Timing and Power Tables

Input pin capacitances:

Input Pins	Rise Cap (pF)	Fall Cap (pF)	Average Cap (pF)
A	0.0024	0.0023	0.0023

Transition Time Table: (20% and 80% of V_{DD})

(i) Output Rise Transitions (in ns) [Input slew vs output capacitance].

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.0169	0.0171	0.0324
10 fF	0.0722	0.0722	0.0776
100 fF	0.6136	0.6136	0.6143

(ii) Output Fall Transitions (in ns) [Input slew vs output capacitance].

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.0172	0.0175	0.0305
10 fF	0.0757	0.0757	0.0819
100 fF	0.6444	0.6444	0.6455

Propagation Delay Time Tables: (50% of input to 50% of output, other inputs held fixed)

(i) Cell Rise Delay (in ns) [Input slew vs output capacitance].

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.0478	0.0661	0.1430
10 fF	0.0886	0.1069	0.1921
100 fF	0.4577	0.4762	0.5624

(ii) Cell Fall Delay (in ns) [Input slew vs output capacitance].

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.0482	0.0693	0.1568
10 fF	0.0970	0.1184	0.2135
100 fF	0.5307	0.5531	0.6516

Static Power: (all possible input combinations)

Condition (A)	Power (nW)
0	0.135
1	0.135

Dynamic Power Table: (Input Time Period = 10 ns)

(i) Rise Power (in nW) [Input slew vs output capacitance].

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	739.3	783.2	793.1
10 fF	649.2	691.7	711
100 fF	765.5	780.5	816.6

(ii) Fall Power (in nW) [Input slew vs output capacitance].

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	1409.8	1461.4	1472.8
10 fF	1335.7	1394.7	1421.1
100 fF	1498.3	1536.6	1597.9

2 Cell: dfxtp

2.1 Circuit Diagram and Transistor Sizing

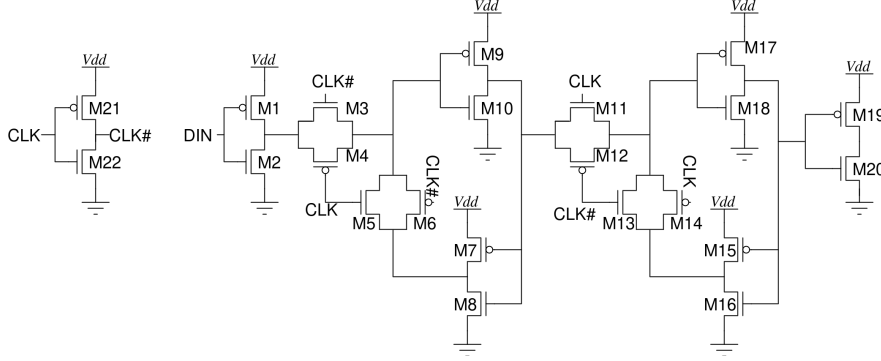


Figure 8: Circuit diagram of dfxtp

Table 2: MOSFET widths and lengths for dfxtp

Device	Type	W (μm)	L (μm)
M1	PMOS	1.26	0.15
M2	NMOS	0.42	0.15
M3	NMOS	0.42	0.15
M4	PMOS	1.23	0.15
M5	NMOS	0.42	0.15
M6	PMOS	1.26	0.15
M7	PMOS	1.26	0.15
M8	NMOS	0.42	0.15
M9	PMOS	1.26	0.15
M10	NMOS	0.42	0.15
M11	NMOS	0.42	0.15
M12	PMOS	1.26	0.15
M13	NMOS	0.42	0.15
M14	PMOS	1.26	0.15
M15	PMOS	1.26	0.15
M16	NMOS	0.42	0.15
M17	PMOS	1.26	0.15
M18	NMOS	0.42	0.15
M19	PMOS	1.26	0.15
M20	NMOS	0.42	0.15
M21	PMOS	1.26	0.15
M22	NMOS	0.42	0.15

2.2 Layout

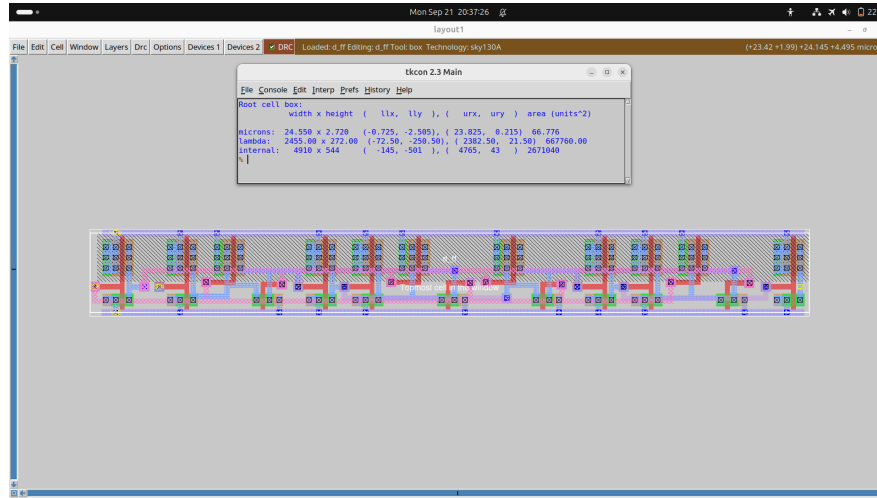


Figure 9: Layout of dfxtp in Magic

Cell area: $\text{WIDTH} \times \text{HEIGHT} = 24.55\mu\text{m} \times 2.72\mu\text{m}$

2.3 PEX Netlist

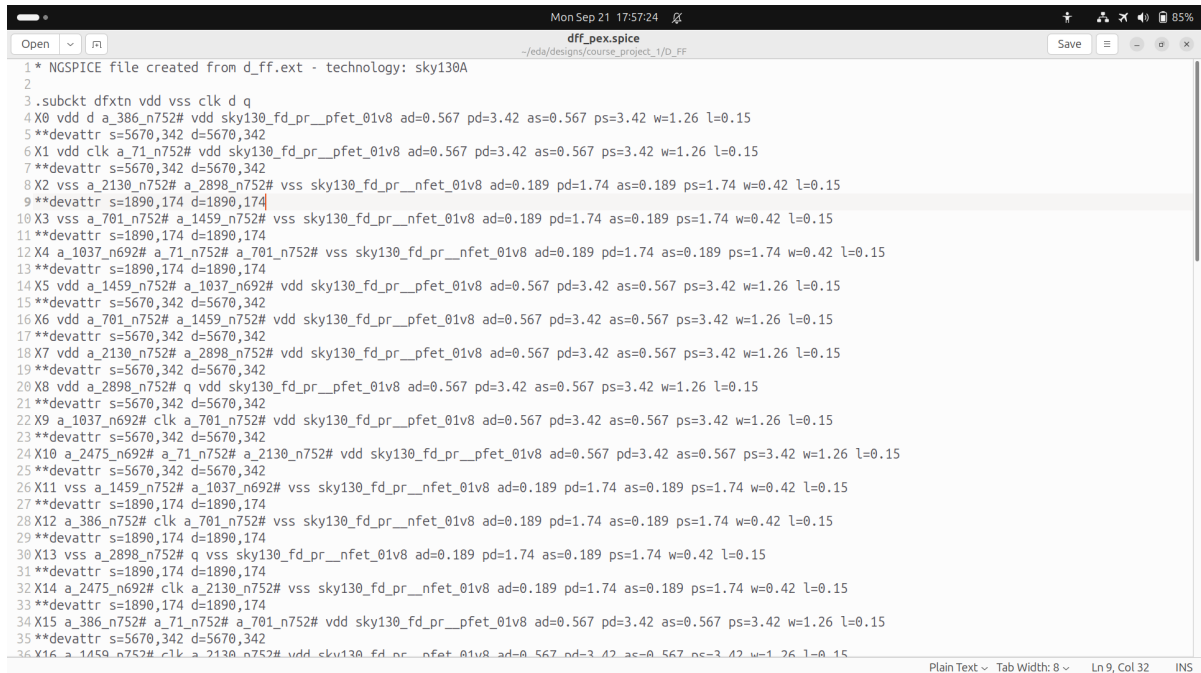


Figure 10: PEX netlist Part 1 of dfxtp

Figure 11: PEX netlist Part 2 of dfxtp

Figure 12: PEX netlist Part 3 of dfxtp

2.4 DRC and LVS

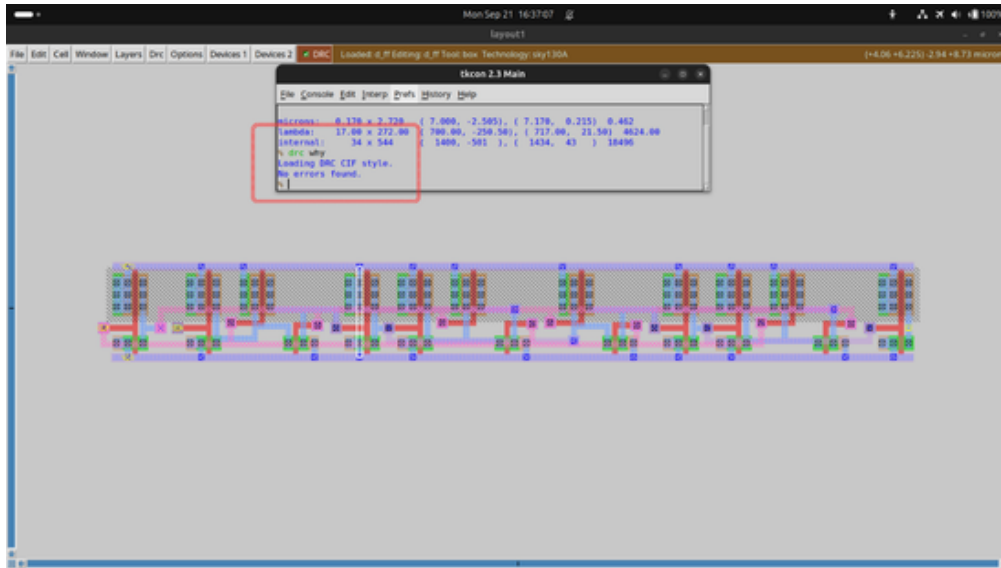


Figure 13: DRC clean screenshot for dfxtp

```

comp.out
Comparison output logged to file comp.out
Logging to file "comp.out" enabled
Contents of circuit 1: Circuit: 'd_ff'
Circuit d_ff contains 22 device instances.
  Class: sky130_fd_pr_nfet_01v8 instances: 11
  Class: sky130_fd_pr_pfet_01v8 instances: 11
Circuit contains 14 nets.
Contents of circuit 2: Circuit: 'd_ff'
Circuit d_ff contains 22 device instances.
  Class: sky130_fd_pr_nfet_01v8 instances: 11
  Class: sky130_fd_pr_pfet_01v8 instances: 11
Circuit contains 13 nets.

Circuit 1 contains 22 devices, Circuit 2 contains 22 devices.
Circuit 1 contains 13 nets, Circuit 2 contains 13 nets.

Netlists match uniquely.
Result: Circuits match uniquely.
Logging to file "comp.out" disabled
LVS Done.
meet@ubuntu:~/eda/designs/course_project_1$

```

Figure 14: LVS clean screenshot for dfxtp

2.5 Simulation Waveforms

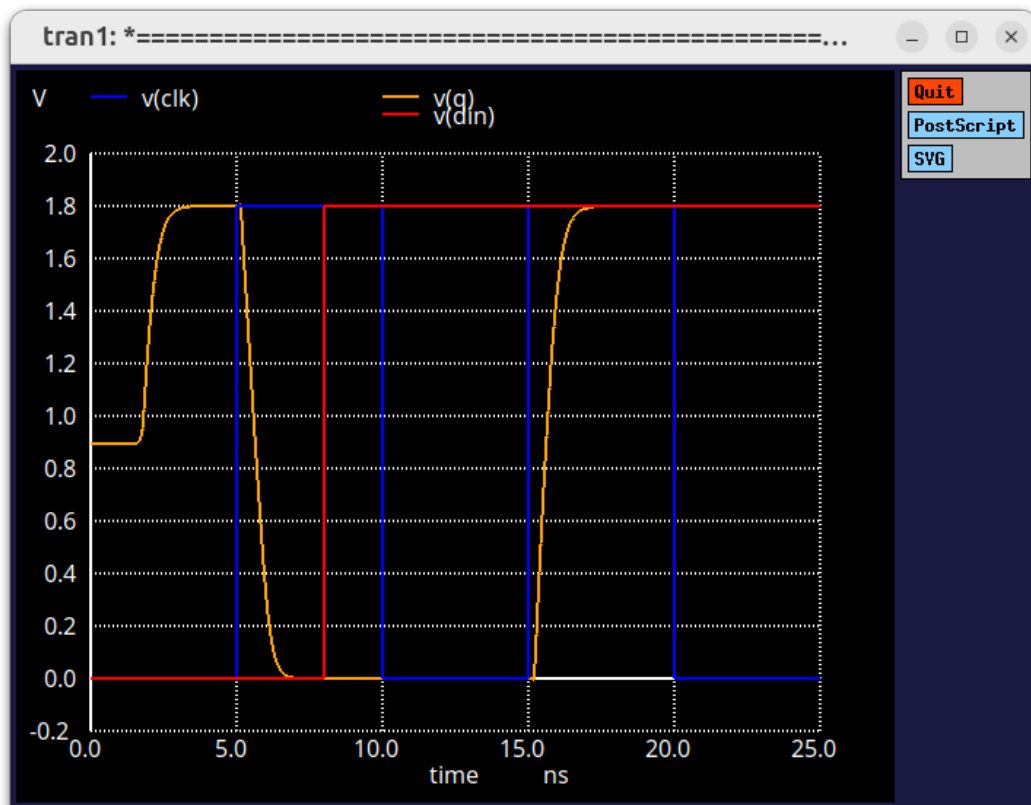


Figure 15: Propagation delay waveform (input, output) for dfxtp

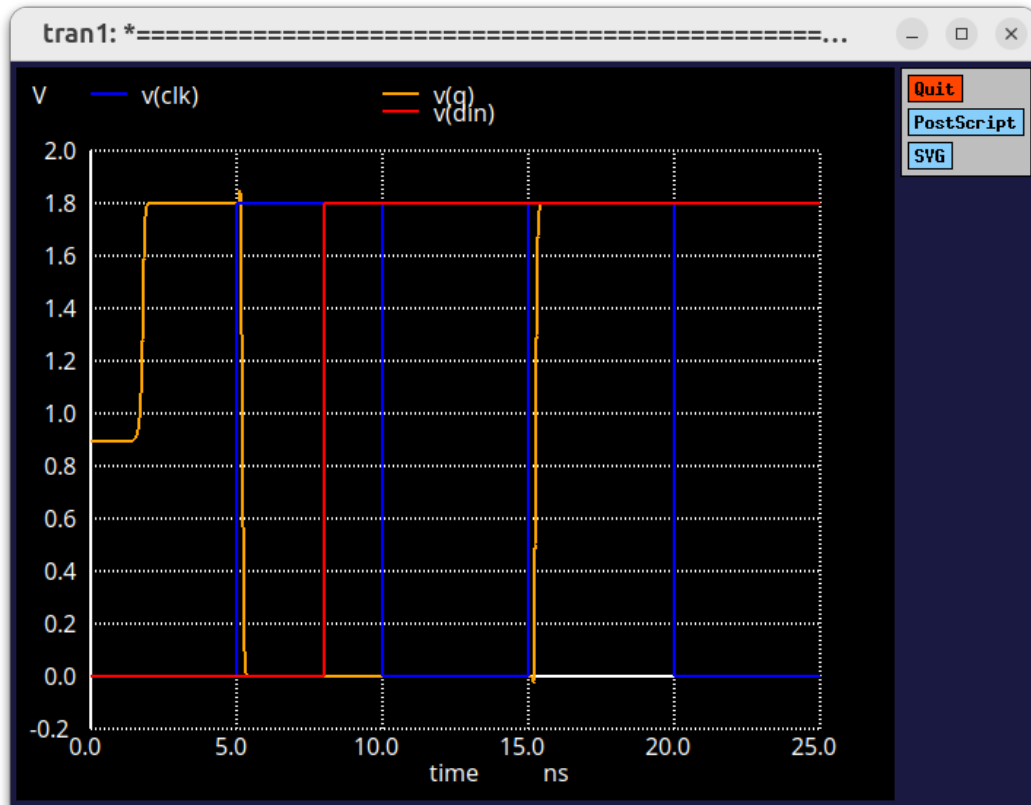


Figure 16: Rise/fall transition time waveform for dfxtp

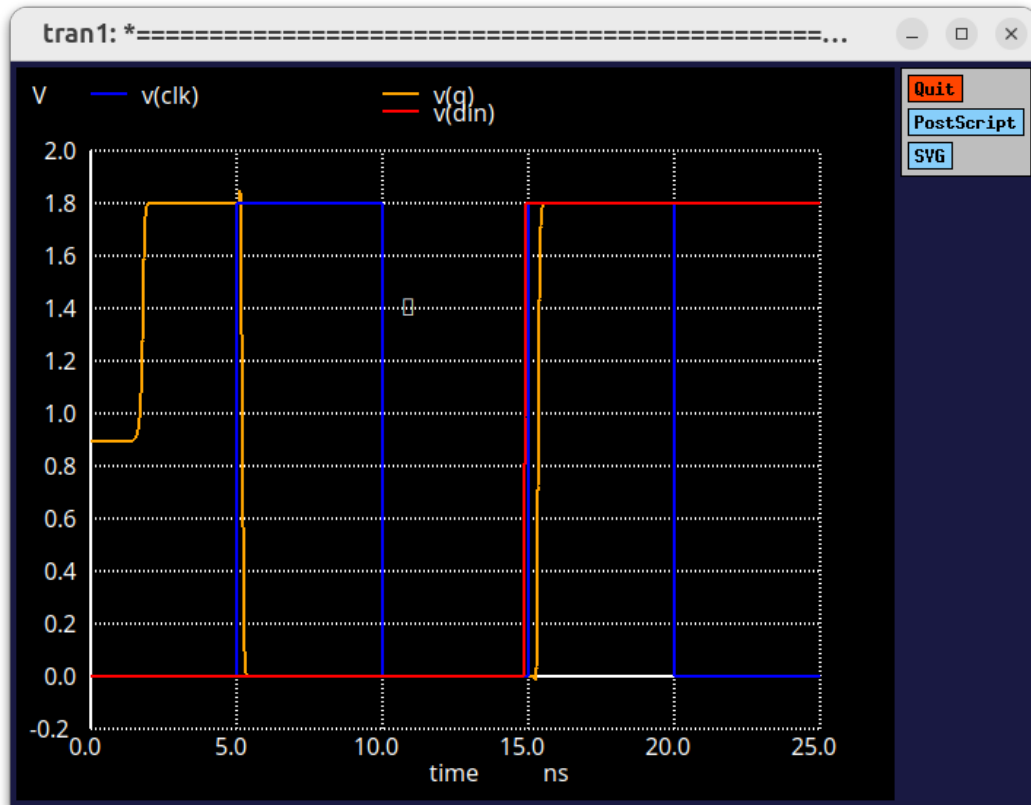


Figure 17: Set-up time waveform (data, clock, output) for dfxtp

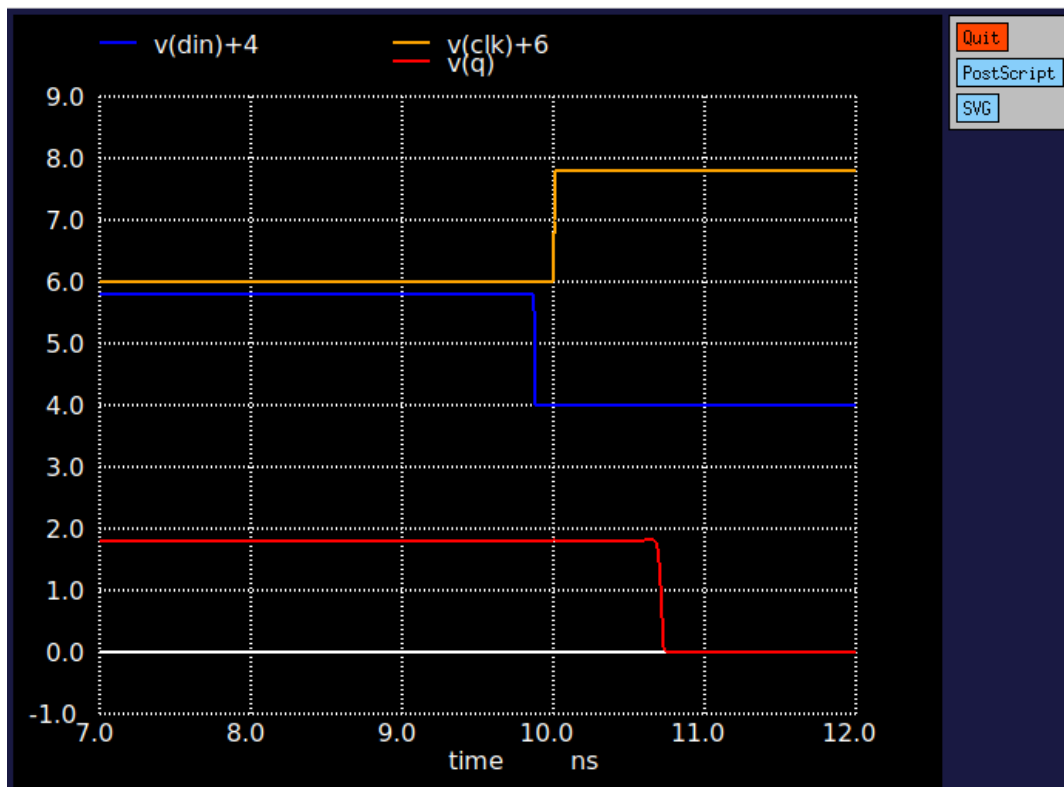


Figure 18: Hold time waveform (data, clock, output) for dfxtp

2.6 Timing and Power Tables

Input pin capacitances:

Input Pins	Rise Cap (pF)	Fall Cap (pF)	Average Cap (pF)
D	0.00222	0.00216	0.00219
CLK	0.00841	0.00839	0.00840

Set-up Time Table:

(i) Rise Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0.121	0
1000 ps	0.782	0.165

(ii) Fall Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0.130	0
1000 ps	0.802	0.178

Hold Time Table:

(i) Rise Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	-0.1209	-0.0102
1000 ps	-0.2565	-0.1645

(ii) Fall Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	-0.1295	-0.2105
1000 ps	-0.3065	-0.3755

Transition Time Table (for Q output, CLK has minimum slew of 10 ps): (20% and 80% of V_{DD})

(i) Output Rise Transitions (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.0203	0.0204	0.0205
10 fF	0.709	0.0712	0.0713
100 fF	0.5975	0.5977	0.5979

(ii) **Output Fall Transitions (in ns) [Input slew vs output capacitance].**

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.0179	0.01799	0.01825
10 fF	0.7530	0.07534	0.07538
100 fF	0.64083	0.64087	0.64092

CLK-to-Q Delay Time Table: (50% of CLK to 50% of output)

(i) **Cell Rise Delay (in ns) [Input slew vs output capacitance].**

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.1928	0.1930	0.1932
10 fF	0.2421	0.2434	0.2436
100 fF	0.6017	0.6067	0.6072

(ii) **Cell Fall Delay (in ns) [Input slew vs output capacitance].**

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.13778	0.13794	0.1382
10 fF	0.19653	0.19654	0.19717
100 fF	0.63343	0.63349	0.63356

Static Power: (all possible input combinations of CLK and D)

Condition (CLK, D)	Power (nW)
00	67778.58
01	0.3676506
10	0.4729598
11	0.4729599

Dynamic Power Table: (Clock Time Period = 10 ns)

(i) **Rise Power (in nW) [Input slew vs output capacitance].**

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	15285	14714	14767
10 fF	18954	18385	18447
100 fF	47578	47009	47073

(ii) **Fall Power (in nW) [Input slew vs output capacitance].**

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	11077	11077	11077
10 fF	10765	10764	10764
100 fF	11290	11290	11290

3 Cell: nand2b

3.1 Circuit Diagram and Transistor Sizing

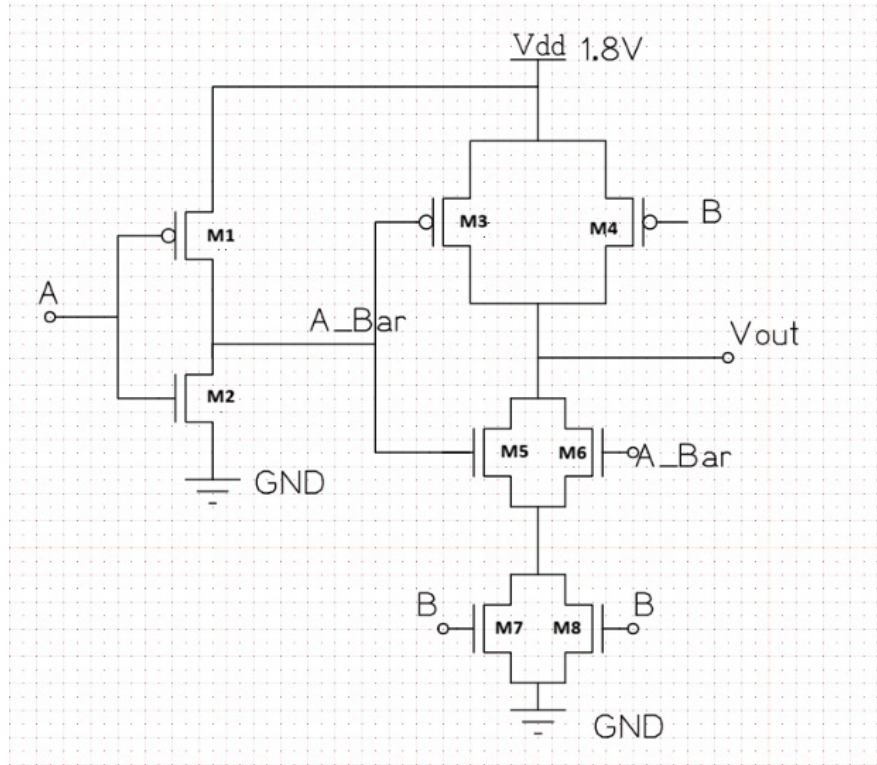


Figure 19: Circuit diagram of nand2b

- To preserve the same rise time t_r and fall time t_f as the reference circuit, the resistance contributed by each series NMOS must be halved, which means each transistor's width must be doubled, and hence its aspect ratio doubles as well.
- However, doubling the width of a single transistor would also double its height in layout, which would violate the fixed cell height constraint of $2.72\ \mu\text{m}$. To resolve this, the fingering technique is used.
- In the fingering technique, instead of implementing one NMOS of width $2W$ as a single device, it is split into two NMOS transistors of width W each, connected in parallel. This achieves the same effective width (and hence the same equivalent resistance) while keeping the cell height fixed at $2.72\ \mu\text{m}$.

Table 3: MOSFET widths and lengths for nand2b

Device	Type	W (μm)	L (μm)
M1	PMOS	1.23	0.15
M2	NMOS	0.42	0.15
M3	PMOS	1.23	0.15
M4	PMOS	1.23	0.15
M5	NMOS	0.42	0.15
M6	NMOS	0.42	0.15
M7	NMOS	0.42	0.15
M8	NMOS	0.42	0.15

3.2 Layout

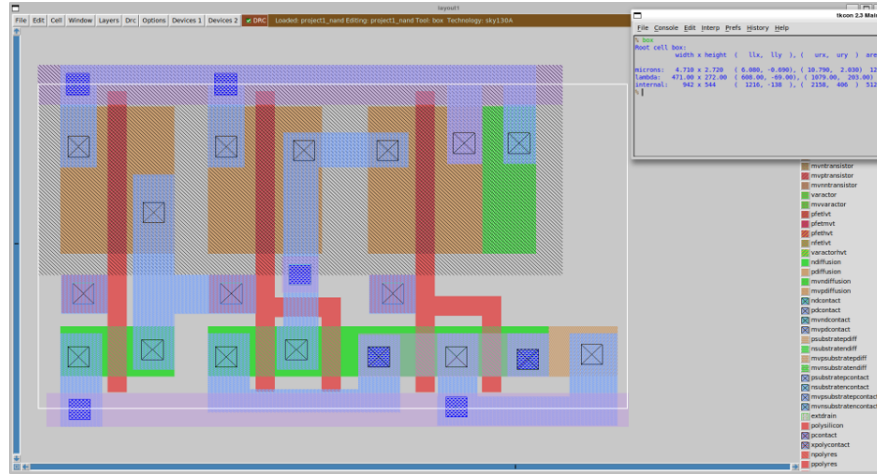
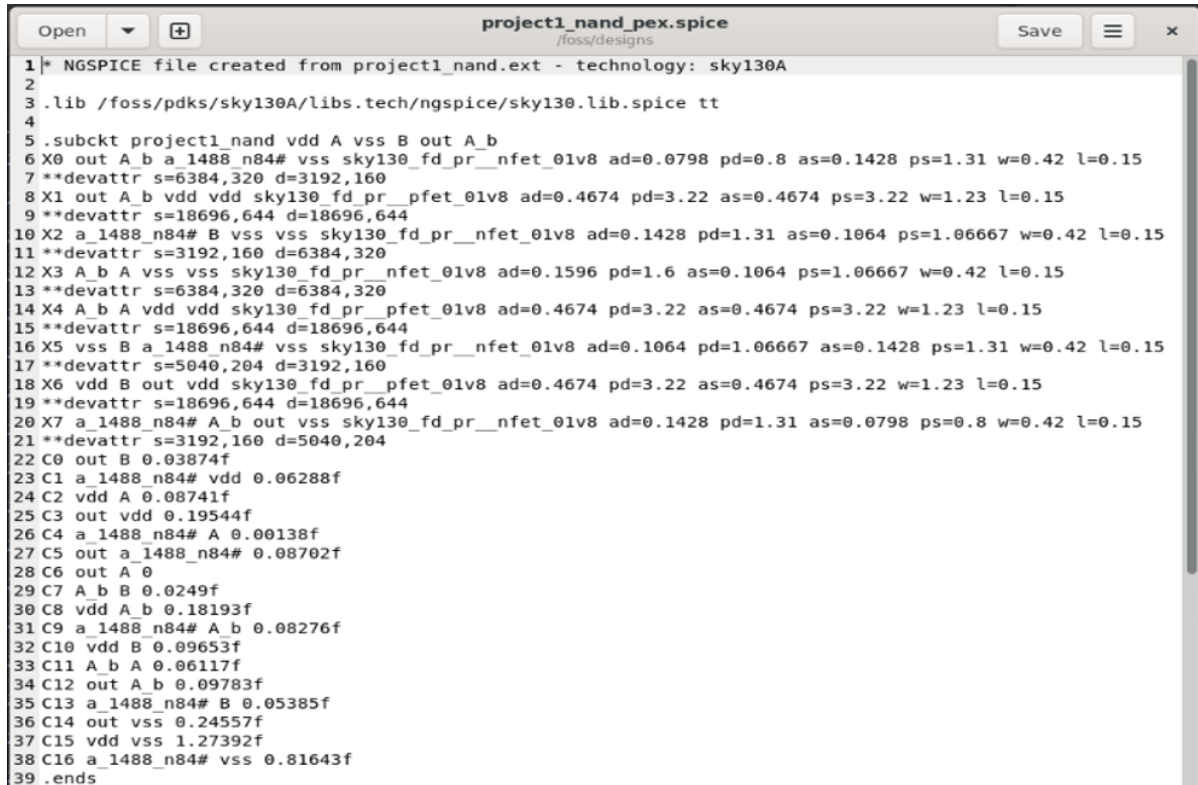


Figure 20: Layout of nand2b in Magic

Cell area: $\text{WIDTH} \times \text{HEIGHT} = 4.71 \mu\text{m} \times 2.72 \mu\text{m}$

3.3 PEX Netlist



```

1 NGSPICE file created from project1_nand.ext - technology: sky130A
2
3 .lib /foss/pdks/sky130A/libs.tech/ngspice/sky130.lib.spice tt
4
5 .subckt project1_nand vdd A vss B out A b
6 X0 out A_b a_1488_n84# vss sky130_fd_pr__nfet_01v8 ad=0.0798 pd=0.8 as=0.1428 ps=1.31 w=0.42 l=0.15
7 **devattr s=6384,320 d=3192,160
8 X1 out A_b vdd sky130_fd_pr__pfet_01v8 ad=0.4674 pd=3.22 as=0.4674 ps=3.22 w=1.23 l=0.15
9 **devattr s=18696,644 d=18696,644
10 X2 a_1488_n84# B vss sky130_fd_pr__nfet_01v8 ad=0.1428 pd=1.31 as=0.1064 ps=1.06667 w=0.42 l=0.15
11 **devattr s=3192,160 d=6384,320
12 X3 A_b A vss sky130_fd_pr__nfet_01v8 ad=0.1596 pd=1.6 as=0.1064 ps=1.06667 w=0.42 l=0.15
13 **devattr s=6384,320 d=6384,320
14 X4 A_b A vdd sky130_fd_pr__pfet_01v8 ad=0.4674 pd=3.22 as=0.4674 ps=3.22 w=1.23 l=0.15
15 **devattr s=18696,644 d=18696,644
16 X5 vss B a_1488_n84# vss sky130_fd_pr__nfet_01v8 ad=0.1064 pd=1.06667 as=0.1428 ps=1.31 w=0.42 l=0.15
17 **devattr s=5040,204 d=3192,160
18 X6 vdd B out vdd sky130_fd_pr__pfet_01v8 ad=0.4674 pd=3.22 as=0.4674 ps=3.22 w=1.23 l=0.15
19 **devattr s=18696,644 d=18696,644
20 X7 a_1488_n84# A b out vss sky130_fd_pr__nfet_01v8 ad=0.1428 pd=1.31 as=0.0798 ps=0.8 w=0.42 l=0.15
21 **devattr s=3192,160 d=5040,204
22 C0 out B 0.03874f
23 C1 a_1488_n84# vdd 0.06288f
24 C2 vdd A 0.08741f
25 C3 out vdd 0.19544f
26 C4 a_1488_n84# A 0.00138f
27 C5 out a_1488_n84# 0.08702f
28 C6 out A 0
29 C7 A_b B 0.0249f
30 C8 vdd A_b 0.18193f
31 C9 a_1488_n84# A_b 0.08276f
32 C10 vdd B 0.09653f
33 C11 A_b A 0.06117f
34 C12 out A_b 0.09783f
35 C13 a_1488_n84# B 0.05385f
36 C14 out vss 0.24557f
37 C15 vdd vss 1.27392f
38 C16 a_1488_n84# vss 0.81643f
39 .ends

```

Figure 21: Screenshot of the PEX netlist of nand2b

3.4 DRC and LVS

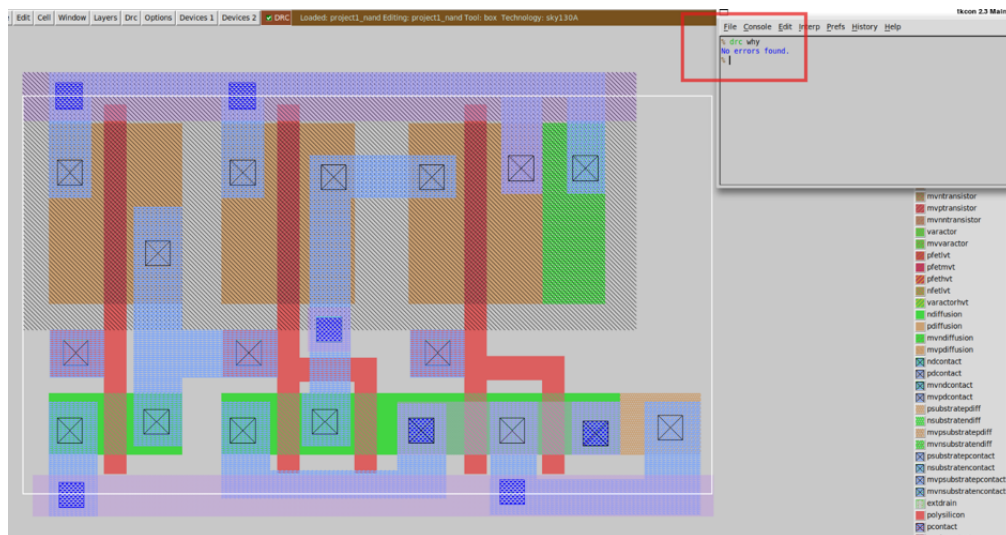


Figure 22: DRC clean screenshot for nand2b

```
Circuit nand_abar_b contains 6 device instances.
  Class: skyl30_fd_pr_pfet_01v8 instances:  3
  Class: skyl30_fd_pr_nfet_01v8 instances:  3
Circuit contains 7 nets.

Circuit was modified by parallel/series device merging.
New circuit summary:

Contents of circuit 1:  Circuit: 'project1_nand'
Circuit project1_nand contains 6 device instances.
  Class: skyl30_fd_pr_pfet_01v8 instances:  3
  Class: skyl30_fd_pr_nfet_01v8 instances:  3
Circuit contains 7 nets.
Contents of circuit 2:  Circuit: 'nand_abar_b'
Circuit nand_abar_b contains 6 device instances.
  Class: skyl30_fd_pr_pfet_01v8 instances:  3
  Class: skyl30_fd_pr_nfet_01v8 instances:  3
Circuit contains 7 nets.

Circuit 1 contains 6 devices, Circuit 2 contains 6 devices.
Circuit 1 contains 7 nets,    Circuit 2 contains 7 nets.

Final result:
Circuits match uniquely.
Property errors were found.

The following cells had property errors:
  project1_nand

Logging to file "comp.out" disabled
LVS Done.
/foss/designs/project1/project1_nand > █
```

Figure 23: LVS clean screenshot for nand2b

3.5 Simulation Waveforms

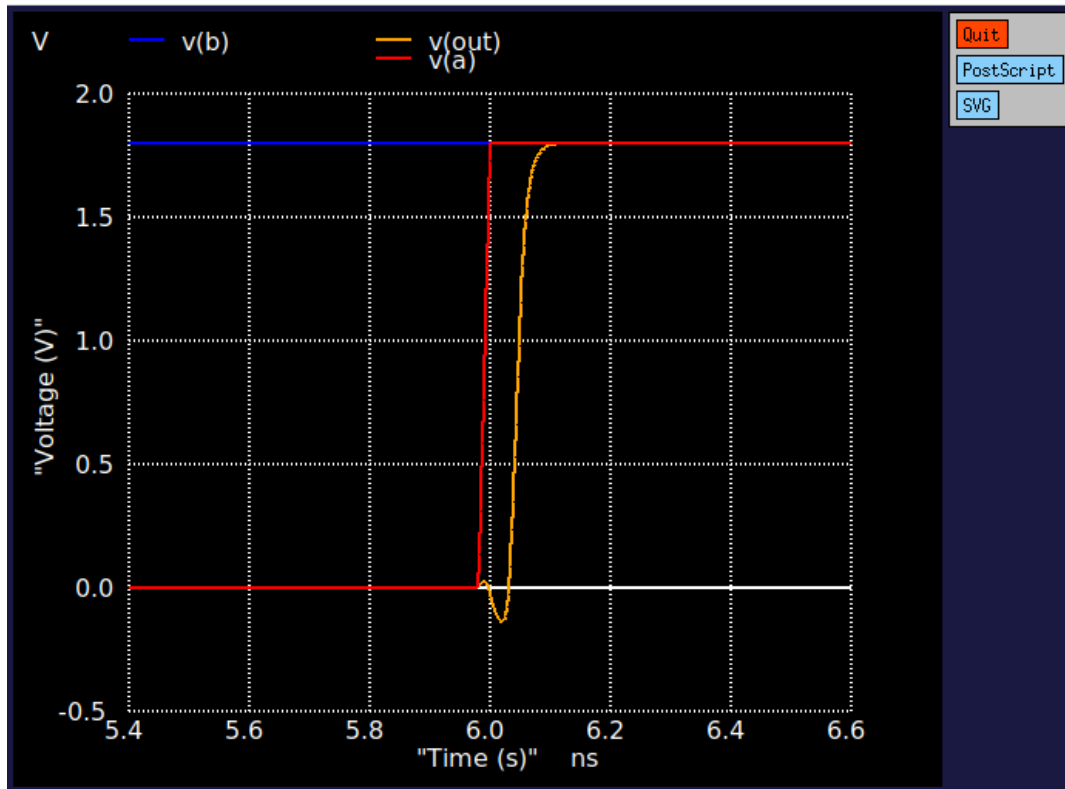


Figure 24: Propagation delay waveform (input, output) for nand2b

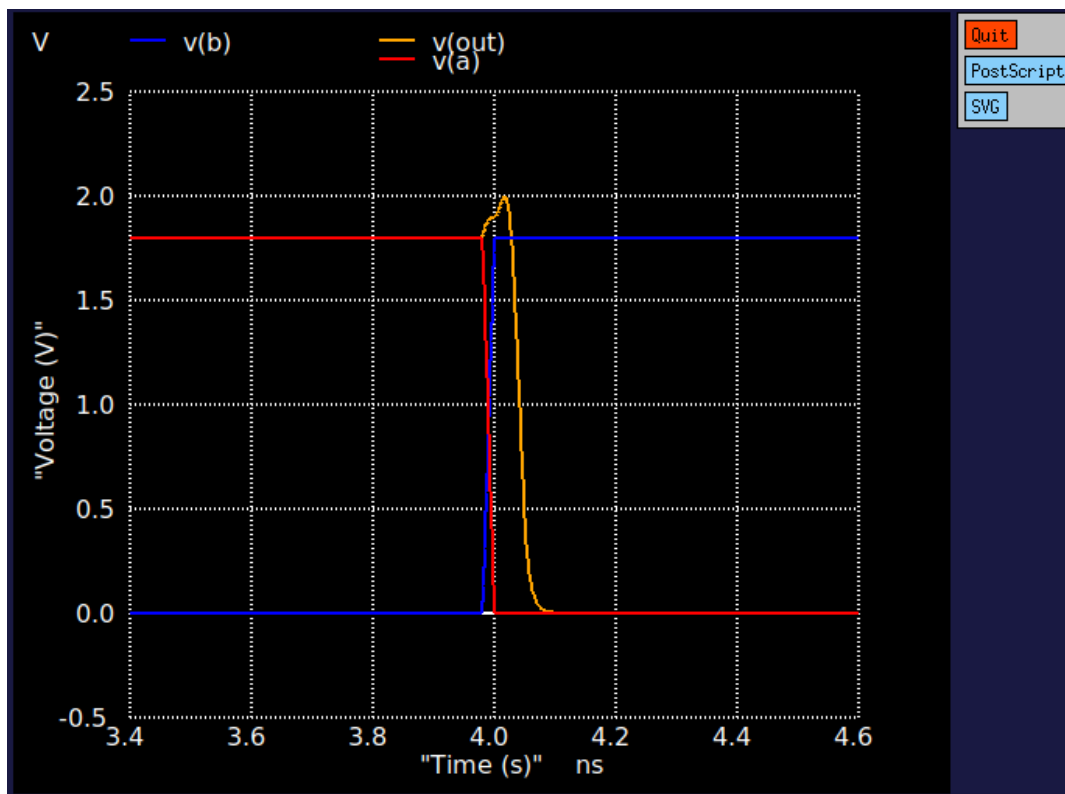


Figure 25: Rise/fall transition time waveform for nand2b

3.6 Timing and Power Tables

Input pin capacitances:

Input Pins	Rise Cap (pF)	Fall Cap (pF)	Average Cap (pF)
A	0.00225697	0.002169711	0.002213341
B	0.002921491	0.002865751	0.002893621

Transition Time Table: (20% and 80% of V_{DD})

(i) Output Rise Transitions (in ns) [Input slew vs output capacitance].

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.02198	0.02208	0.0373
10 fF	0.07724	0.07725	0.08225
100 fF	0.61861	0.61861	0.61912

Related pin B: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.02958	0.03219	0.1132
10 fF	0.08625	0.08621	0.19255
100 fF	0.62807	0.62807	0.63745

(ii) Output Fall Transitions (in ns) [Input slew vs output capacitance].

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.01975	0.01995	0.03353
10 fF	0.06826	0.06829	0.07577
100 fF	0.53635	0.53635	0.53738

Related pin B: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.01824	0.02060	0.079
10 fF	0.06793	0.06802	0.137
100 fF	0.53635	0.53635	0.54708

Propagation Delay Time Tables: (50% of input to 50% of output, other inputs held fixed)

(i) Cell Rise Delay (in ns) [Input slew vs output capacitance].

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	56.95	75.55	165.10
10 fF	98.67	117.11	211.97
100 fF	468.50	487.27	582.86

Related pin B: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	47.81	66.14	143.03
10 fF	88.69	106.97	192.15
100 fF	457.76	476.24	562.42

(ii) Cell Fall Delay (in ns) [Input slew vs output capacitance].

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	51.23	72.68	166.20
10 fF	91.16	112.81	213.66
100 fF	443.88	466.44	570.51

Related pin B: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	48.22	69.37	156.80
10 fF	97.07	118.47	213.57
100 fF	530.70	553.18	651.61

Static Power: (all possible input combinations)

Condition (AB)	Power (nW)
00	0.019
01	0.138
10	0.267
11	0.138

Dynamic Power Table: (Input Time Period = 10 ns)

(i) Rise Power (in nW) [Input slew vs output capacitance].

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	1280	1176	1296
10 fF	5232	5121	5175
100 fF	41731	41639	41731

Related pin B: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	1843	1756	2025
10 fF	5698	5636	5785
100 fF	42147	42121	42074

(ii) **Fall Power (in nW) [Input slew vs output capacitance].**

Related pin A: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	2275	2190	2379
10 fF	2335	2260	2424
100 fF	2354	2297	2506

Related pin B: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	621	531	577
10 fF	640	594	546
100 fF	646	638	598

4 Contribution of Team Members

Member	Roll No.	Contribution
Meet Jain	26M1164	DFF layout, Sequential lib file, Charlib, Report
Rajas Tatwawadi	26M1161	BUF, NAND layout, Combinational lib files, Report
Abhineet Singh	26M1191	Combinational and Sequential lib files
Uday Gupta	26X0040	BUF, NAND layout, Charlib, Report
